

National Contest
Team Selection Test - DAY 1
Date: March 19, 2025

Duration: 4 hours and 30 minutes

Problems are arranged by relative difficulty

Problem 7.3.1 (Sol P65)

Given n points, where $n \geq 4$, labeled $1, 2, \dots, n$, placed consecutively on a circle. Two players, A and B , play a game in which they alternately draw a chord whose endpoints have the same parity.

It is not allowed for chords to intersect (not even at endpoints). Player A starts the game. At the end of the game, the winner is the player who drew the last chord.

Question : Which player has a winning strategy?

Hint: 93

Problem 7.3.2 (Sol P66)

Let ABC be a non-isosceles triangle such that $AB < AC$, and let I be the incenter of the triangle. Let AT be the external angle bisector of $\angle BAC$, with $T \in (BC)$. Consider the point K on the perpendicular bisector of BC such that $IT \perp IK$. Let O be the circumcenter of triangle ABC , and let R be its circumradius.

Prove that $OK = 3R$.

Problem 7.3.3 (Sol P68)

Let $a_1, \dots, a_k$ be strictly positive integers, and let $f(x)$ be a non-constant polynomial with integer coefficients. Prove that there exists an infinite set A of prime numbers such that for every $p \in A$, the expression

$$f(n) + a_1^n + \dots + a_k^n$$

is divisible by p for infinitely many values of n .

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National Contest
Team Selection Test - DAY 2
Date: March 20, 2025

Duration: 4 hours and 30 minutes

Problems are arranged by relative difficulty

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Problem 7.4.4 (Sol P69)

A n -tuple $(a_1, a_2, \dots, a_n)$ of positive integers is called *admissible*, with $n \in \mathbb{N}^*$, if its elements are pairwise distinct and satisfy the following condition : for all $i, j \in \{1, \dots, n\}$, there exists $k \in \{1, \dots, n\}$ such that $a_i \equiv -2a_k \pmod{a_j}$. Determine all admissible n -tuples.

Problem 7.4.5 (Sol P70)

Let $\mathbb{R}_{>0}$ be the set of strictly positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ satisfying

$$x + f(yf(x) + 1) = xf(x + y) + yf(yf(x))$$

for all strictly positive real numbers x and y .

Problem 7.4.6 (Sol P72)

In a country, cities are connected by roads. Each road is classified into one of four categories : a road managed by the State, or by one of the three companies $\{A, B, C\}$.

The country is said to be **happy** if the following two conditions are satisfied :

1. In any city where exactly three roads meet, these roads must either all be managed by the State, or be managed separately by companies A, B, and C.
2. There exists at least one road that is not managed by the State.

The country is composed of exactly of a cities where exactly four roads meet, and b cities where exactly three roads meet. Find the smallest real number λ such that if

$a > \lambda b$ then the country is **happy**